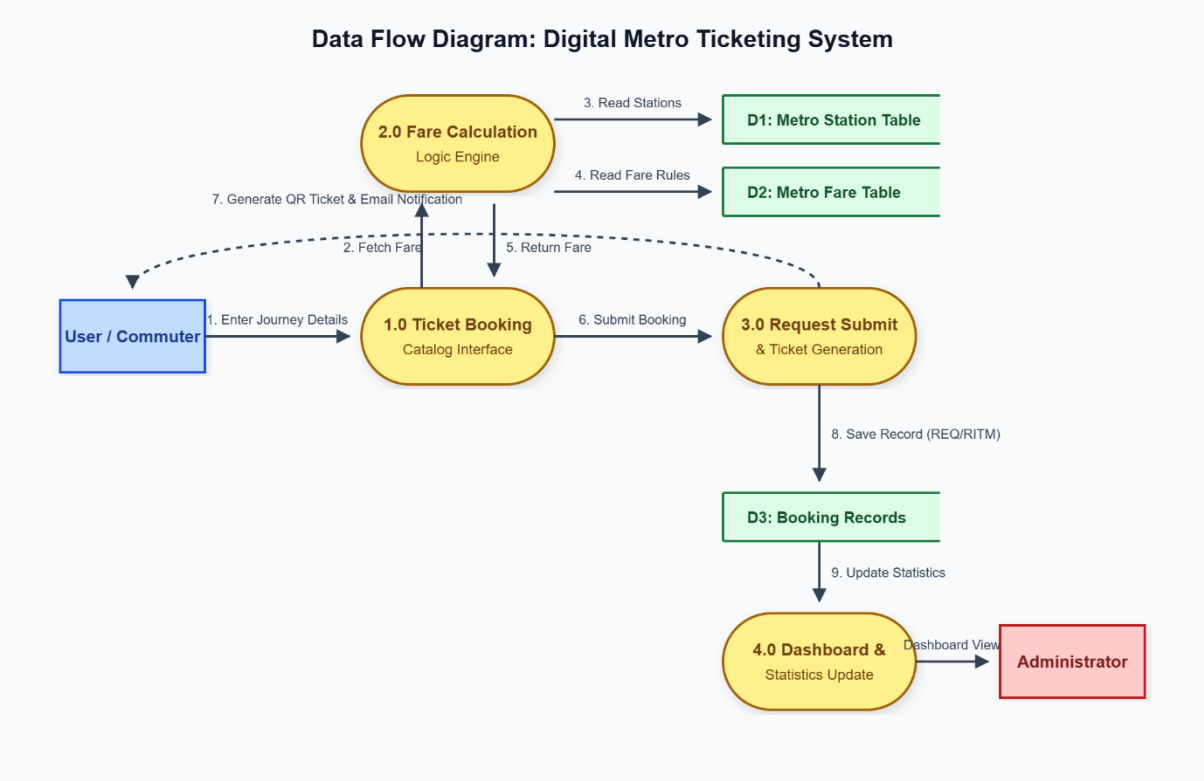
Project Design Phase-II

Data Flow Diagram & User Stories

Date	28-09-26
Team ID	23KD1A4230
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

The Data Flow Diagram (DFD) below maps the logical sequence of the Metro Ticket Booking process as referenced in the system architecture. It illustrates how a commuter's journey details are processed through the fare calculation engine, validated against the station databases, and ultimately transformed into a generated QR ticket and administrative reports.



User Stories

The following user stories have been mapped according to the system's data flow, outlining requirements across the application's lifecycle from the perspective of both the commuter and the administrator.

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Priority	Release
Commuter	Booking	USN-1	As a commuter, I want to select my source and destination so I can begin my travel booking.	Dropdowns populate with all active metro stations.	High	Sprint-1
Commuter	Fare Logic	USN-2	As a commuter, I want the system to calculate my fare instantly so I know the exact cost.	The fare field auto-populates upon station selection.	High	Sprint-1
Commuter	Payment	USN-3	As a commuter, I want to select my preferred payment method so I can complete my purchase.	Options for digital wallets, UPI, and cards are available.	High	Sprint-1
Commuter	Ticketing	USN-4	As a commuter, I want a digital QR code upon submission so I can scan it at the gate.	A scannable QR widget renders successfully on screen.	High	Sprint-1
Commuter	Notification	USN-5	As a commuter, I want an email confirmation so I have a reliable backup of my ticket.	Automated email triggers with booking details attached.	Medium	Sprint-1
Admin	Dashboard	USN-6	As an admin, I want to view live daily booking statistics so I can monitor station traffic.	Dashboard widgets update accurately with current bookings.	High	Sprint-2
Admin	Station Data	USN-7	As an admin, I want to add new metro stations to the database so the routing stays current.	Station details table allows create/edit operations.	Medium	Sprint-2
Admin	Fare Rules	USN-8	As an admin, I want to update base fare pricing so calculations match new organizational rates.	Changes to the fare table instantly apply to new bookings.	High	Sprint-2
Commuter	History	USN-9	As a commuter, I want to view my past digital tickets so I can easily track my travel history.	A 'My Tickets' section displays all previous travel records.	Low	Sprint-3
Admin	Support	USN-10	As an admin, I want to track failed booking submissions so I can troubleshoot commuter issues.	Failed REQ/RITM transactions are flagged in the backend.	Medium	Sprint-3